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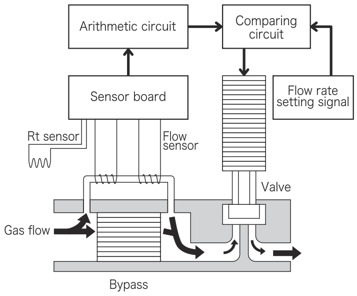
GAS | Products For Gas Applications

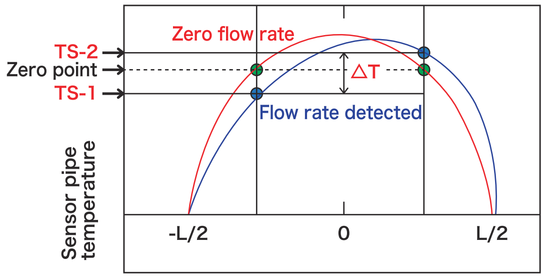
Deploying advance technologies, LINTEC has been focusing on developing high-performance mass flow controllers for next generation semiconductor industries. By addressing growing market demands for products with higher accuracy, reliability, speed, and higher level of ultra-cleanliness, we are ready to serve more industries in the near future.

Introduction to Mass Flow Controller

Mass flow controller is an apparatus used to control mass flow of various gases and liquids with high precision. Mass flow controller is used extensively in applications for semiconductor, LCD, and the manufacturing of electric devices. It is able to achieve high-accuracy control because gas and liquid flow rates are measured not in volume but in mass, thus nullifying the dependence on temperature or pressure. Mass flow controller is composed of a sensor, a valve, and an electronic circuit. A small portion of the gas supplied flows separately through the bypass channel and the sensor integrated within the controller. Feed back control is accomplished by comparing the mass flow rate output signal detected by the sensor and the flow rate setting signal, which eventually controls the opening and closing of the valve.



[Inner structure]

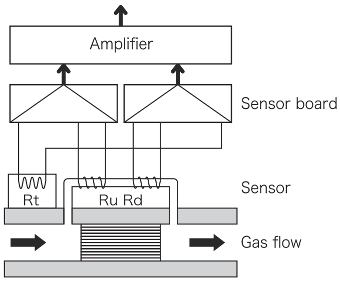


TS-2 - TS-1=ΔT Qm=K ΔT
[Principle for fluid detection]

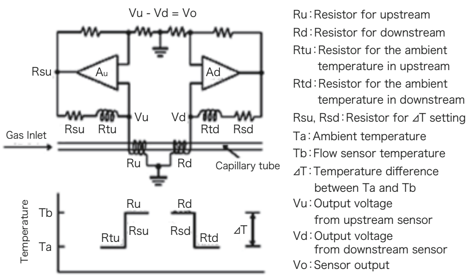
LINTEC's Strength

LINTEC's original ambient temperature-compensated sensor

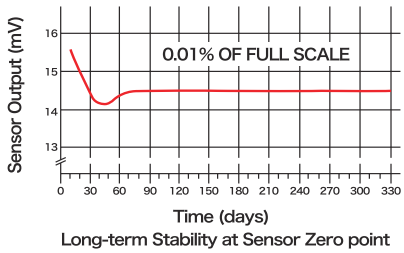
Temperature sensor integrated within our controllers is composed of an ambient temperature sensor (Rt) and two temperature-sensitive winding materials situated upstream (Ru) and downstream (Rd) of the sensor, respectively. Serving as a temperature detector of the sensor, Ru and Rd are controlled by the constant temperature difference of the ambient temperature detected by Rt. When gas flow is not detected, the voltage provided to both Ru and Rd constantly remains at a certain value. In case when gas flow is detected within the system, transfer of heat occurs. As a result, the difference in voltage supplied to both Ru and Rd is generated in order for Ru and Rd to maintain constant temperature difference with the surrounding area. Mass flow signal with high accuracy can be obtained by calculating the ratio of the voltage difference and the flow rate of gas. Since the sensor temperature is adjusted at the time when ambient temperature detection is performed, the sensor output is basically unaffected by changes in ambient temperature. Maintaining lower temperature in comparison with other sensors, our sensor ensures less changes in accuracy during long-term applications and long product life-time.



[Structure of the sensor]



[Principle of LINTEC's flow sensor]



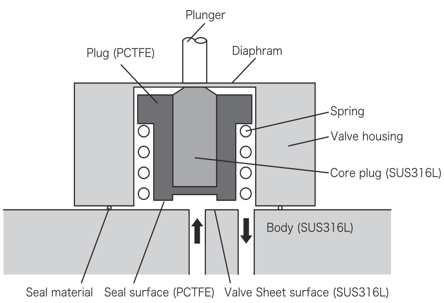
[Changes in the sensitivities of the sensor]

Highly functional upon installation of microprocessor

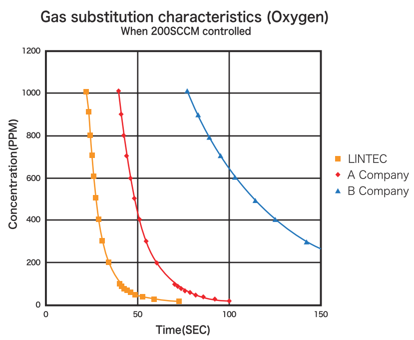
Using microprocessor to memorize output changes and to perform various signal comparisons, error condition can be identified in time by triggering an external alarm alert. Furthermore, automation of flow rate and zero-point adjustment allow stable production of our product with well-maintained quality.

Diaphragm valve with small dead volume

Reduction of dead volume can be acheived by isolating driving actuator and diaphragm valve. This design enhanced gas circulation and is suitable for applications requiring liquefied gas usage. Its prevailing quality against contamination ensures longer product life span.



[Valve structure]



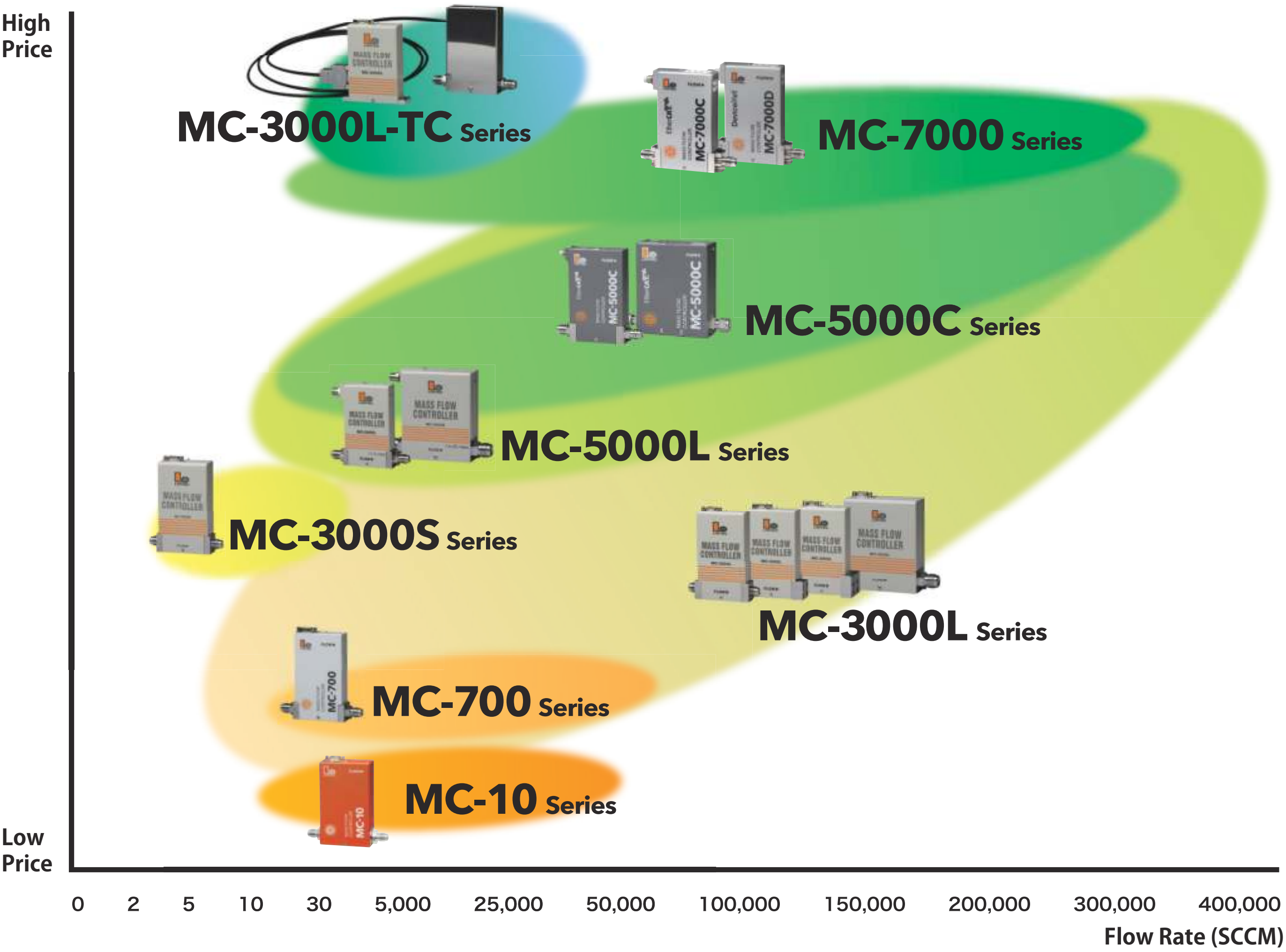
[Gas substitution characteristics]

Product Comparative Table

		Mass Flow Controller			
		EtherCAT® MC-7000C series	DeviceNet® MC-7000D series	EtherCAT® MC-5000C series	DeviceNet® MC-5000L series
Product photo					
Control interface		EtherCAT®	DeviceNet®	EtherCAT®	DeviceNet®
Flow rate in Nitrogen (Full scale)		10SCCM to 20SLM	10SCCM to 20SLM	10SCCM to 300SLM	10SCCM to 200SLM
Accuracy		±1.0% S.P.	±1.0% S.P.	±1.0% S.P.	±1.0% S.P.
Operating temperature		5 to 50℃	5 to 50℃	5 to 50℃	5 to 50℃
Operating differential pressure		50 to 300kPa	50 to 300kPa	50 to 300kPa	50 to 300kPa
Leak integrity		1×10^{-11}Pa・m^3 / sec He	1×10^{-11}Pa・m^3 / sec He	1×10^{-11}Pa・m^3 / sec He	1×10^{-11}Pa・m^3 / sec He
Materials exposed to gas		Stainless steel 316L, PCTFE, PTFE	Stainless steel 316L, PCTFE, PTFE	Stainless steel 316L, PCTFE, PTFE	Stainless steel 316L, PCTFE, PTFE
Seal materials		Stainless steel 316L	Au	Stainless steel 316L	Au
Control valve actuator		Piezoelectric actuator	Piezoelectric actuator	Piezoelectric actuator	Piezoelectric actuator
Options	Surface mount fittings	Optional	Optional	Optional	Optional
	Custom fittings	Optional	Optional	Optional	Optional
	Liquified gas	Optional	Optional	Optional	Optional
	Low full scale	Optional	Optional	Optional	Optional
	Wide range	Optional	Optional	Optional	Optional
	VR function	–	–	–	–
	Digital interface	Default (EtherCAT®)	Default (DeviceNet®)	Default (EtherCAT®)	Default (DeviceNet®)
	2% Cut	Default	Default	Default	Default
Page		7 p.	7 p.	8 p.	9 p.

*The specifications above apply only to standard models. Please see each MFC's separate specification sheet for more details.

Product Lineup



Mass Flow Controller				
High Performance MFC MC-3000L series	Low Diff. Pressure MFC MC-3000S series	Variable Range (VR) MFC MC-700 series	Analog MFC MC-10 series	High Temperature MFC MC-3000L-TC series
				
Analog (0~5V), RS-485	Analog (0~5V), RS-485	Analog (0~5V), RS-485	Analog (0~5V), RS-485	Analog (0~5V), RS-485
5SCCM to 400SLM	2 to 30CCM	10SCCM to 50SLM	10SCCM to 50SLM	10SCCM to 25SLM
±1.0% S.P.	±1.0% F.S.	±1.0% F.S.	±1.0% F.S.	±1.0% F.S.
5 to 50℃	5 to 50℃	15 to 35℃	15 to 35℃	80 to 120℃
50 to 300kPa	8×10^{2}Pa to 1.33×10^{5}Pa	50 to 300kPa	50 to 300kPa	50 to 300kPa
1×10^{-11}Pa・m^3 / sec He	1×10^{-11}Pa・m^3 / sec He	1×10^{-11}Pa・m^3 / sec He	1×10^{-8}Pa・m^3 / sec He	1×10^{-11}Pa・m^3 / sec He
Stainless steel 316L, PCTFE	Stainless steel 316L, Stainless steel 304, PCTFE	Stainless steel 316L, PCTFE, PTFE	Stainless steel 316L, Magnetic stainless steel, PTFE	Stainless steel 316L, PCTFE
Au	Au	Stainless steel 316L	Viton®	Au
Piezoelectric actuator	Piezoelectric actuator	Piezoelectric actuator	Solenoid actuator	Solenoid actuator
Optional	Optional	Optional	–	–
Optional	–	–	–	Optional
Optional	–	–	–	Optional
Optional	–	–	–	–
Optional	–	–	–	–
–	–	Standard	Optional	–
Default (RS-485)	Default (RS-485)	Default (RS-485)	Optional	Default (RS-485)
Optional	Optional	Default	Default	Optional
10 p.	11 p.	12 p.	13 p.	14 p.

Specification Explanation

Flow Rate — Gas flow rate per minute at 0˚C, 1atm (SCCM/SLM)

Accuracy — A degree to which the measured flow rate can deviate from the set point. Can be expressed in % S.P. or % F.S.

Operating Temperature — Temperature range for optimal performance

Operating Differential Pressure — Difference between inlet and outlet pressure necessary for optimal performance. If the differential pressure is too low, the fluid might not flow; if it's too high, the control might become unstable

Leak Integrity — External leak rate

Option Explanation

Surface Mount Fittings — MFC is equipped with surface mount fittings (W seal fittings, C seal fittings etc.)

Custom Fittings — Any fitting types other than VCR or SWL

Liquified Gas (LG) — MFC can perform operations with liquified gas (SiH2Cl2 etc.)

Low Full Scale (SR) — MFC has a low full scale (1 to 4SCCM)

Wide Range (WR) — MFC has a wide flow control range (0.5 to 100% F.S.)

VR Function (R4) — MFC is equipped with variable range (VR) function, which allows users to change full scale and gas type by adjusting 3 rotary switches

Digital Interface (R2) — MFC is equipped with a digital interface

2% Cut (C3) — MFC valve fully closes when set point is lower than 2% F.S.

Fast Response Mass Flow Controller

MC-7000 Series

DeviceNet ODVA CONFORMANT EtherCAT®

■ Outlines

- MC-7000 Series: Excels in less than 200msec fast response performance using thermal sensor

■ Features

- Thermal sensor provides less than 200msec fast response
- LINTEC's unique ambient temperature-compensated sensor allows corrosive gas and liquified gas operations
- Guaranteed ±1.0% S.P. flow rate accuracy
- Equipped with metal seal and piezoelectric actuator
- Available with both DeviceNet™ and EtherCAT® digital interfaces

CE RoHS

Model	MC-7100C	MC-7200C	MC-7100D	MC-7200D
Flow rate in Nitrogen (Full scale)	10 to 5,000SCCM	~20SLM	10 to 5,000SCCM	~20SLM
Flow rate control range	2 to 100% F.S.			
Valve operation mode	Normally open, Normally closed			
Accuracy	100% F.S. to 25% F.S. ±1.0% S.P. 25% F.S. to 2% F.S. ±0.25% F.S.			
Repeatability	±0.2% F.S.			
Response time	0.2sec			
Operating differential pressure	50 to 300kPa	100 to 300kPa (~10SLM) 200 to 300kPa (~20SLM)	50 to 300kPa	100 to 300kPa (~10SLM) 200 to 300kPa (~20SLM)
Withstand pressure	1MPa(G)			
Operating temperature	5 to 50℃ 0 to 80% RH			
Leak integrity	1×10^{-11}Pa・m^3 / sec He			
Materials exposed to gas	Stainless steel 316L, PCTFE, PTFE			
Seal materials	Stainless steel 316L		Au	
Power supply requirement	24VDC			
Mounting position	Free			
Digital interface	EtherCAT®		DeviceNet™	
Control valve actuator	Piezoelectric actuator			
Weight	1.0kg			

EtherCAT® Mass Flow Controller

MC-5000C Series

EtherCAT®

Outlines

- MC-5000C Series: EtherCAT® interface-dedicated mass flow controller

Features

- EtherCAT digital interface in accordance with ETG
- LINTEC's unique ambient temperature-compensated sensor incorporated
- Highly functional upon installation of microprocessor
- Diaphragm valve with small dead volume
- High-speed and high efficiency piezoelectric actuator
- Long-term leak tightness ensured using metal seal
- By using metal case and various filters, stable operation can be achieved through reduction of radio frequency noises and electromagnetic field interferences

CE RoHS

Model		MC-5100C	MC-5200C	MC-5250C	MC-5460C	MC-5470C	MC-5580C
Flow rate in Nitrogen (Full scale)		10 to 5,000SCCM	~20SLM	~50SLM	~150SLM	~200SLM	~300SLM
Flow rate control range		2 to 100% F.S.					
Valve operation mode		Normally open, Normally closed			Normally closed		
Accuracy		100% F.S. to 10% F.S. ±1.0% S.P. 10% F.S. to 2% F.S. ±0.1% F.S.			100% F.S. to 35% F.S. ±1.0% S.P. 35% F.S. to 2% F.S. ±0.35% F.S.	±1.0% F.S.	±2.0% F.S.
Repeatability		±0.2% F.S.			±0.5% F.S.		
Response time		1 sec					
Operating differential pressure		50 to 300kPa	100 to 300kPa	150 to 300kPa	250 to 500kPa	350 to 500kPa	
Withstand pressure		1MPa(G)					
Temperature coefficient	Zero	±0.02% F.S. / ℃			±0.1% F.S. / ℃		
	Span	±0.02% F.S. / ℃	±0.04% F.S. / ℃	±0.08% F.S. / ℃	±0.1% F.S. / ℃		
Operating temperature		5 to 50℃ 0 to 80% RH			10 to 40℃ 0 to 80% RH		
Leak integrity		1×10^{-11}Pa・m^3 / sec He			1×10^{-8}Pa・m^3 / sec He		
Materials exposed to gas		Stainless steel 316L, PCTFE, PTFE			Stainless steel 316L, Stainless steel 304, Teflon®, Viton®		
Seal materials		Stainless steel 316L			Au		
Power supply requirement		24VDC (180mA) ±3%					24VDC (200mA) ±3%
Mounting position		Free					
Digital interface		EtherCAT®					
Control valve actuator		Piezoelectric actuator					
Weight		1.0kg			1.4kg		

DeviceNet™ Mass Flow Controller

MC-5000L Series

Outlines

- MC-5000L Series: DeviceNet™ interface-dedicated mass flow controller in accordance with SEMI

Features

- Widely-used sensor-actuator Networks DeviceNet™(ODVA) interface
- Top or side communication connectors installation available for selection
- LINTEC's unique ambient temperature-compensated sensor incorporated
- Highly functional upon installation of microprocessor
- Diaphragm valve with small dead volume
- High-speed and high performance piezoelectric actuator
- Long-term leak tightness ensured using metal seal
- By using metal case and various filters, stable operation can be achieved through reduction of radio frequency noises and electromagnetic field interferences

CE RoHS

Model	MC-5100L	MC-5200L	MC-5250L	MC-5460L	MC-5470L
Flow rate in Nitrogen (Full scale)	10 to 5,000SCCM	~20SLM	~50SLM	~150SLM	~200SLM
Flow rate control range	2 to 100% F.S.				
Valve operation mode	Normally open, Normally closed			Normally closed	
Accuracy	100% F.S. to 10% F.S. ±1.0% S.P. 10% F.S. to 2% F.S. ±0.1% F.S.			100% F.S. to 35% F.S. ±1.0% S.P. 35% F.S. to 2% F.S. ±0.35% F.S.	±1.0% F.S.
Repeatability	±0.1% F.S.			±0.5% F.S.	
Response time	1 sec				
Operating differential pressure	50 to 300kPa	Normally open valve 50 to 300kPa (~10SLM) 100 to 300kPa (~20SLM) Normally closed valve 100 to 300kPa (~10SLM) 200 to 300kPa (~20SLM)	100 to 300kPa (~30SLM) 150 to 300kPa (~50SLM)	250 to 500kPa	350 to 500kPa
Withstand pressure	1MPa(G)				
Operating temperature	5 to 50℃ 0 to 80% RH				
Leak integrity	1×10^{-11}Pa・m^3 / sec He				
Materials exposed to gas	Stainless steel 316L, PCTFE, PTFE			Stainless steel 316L, Stainless steel 304, Teflon®, Viton®	
Seal materials	Au				
Power supply requirement	11 to 25VDC (5VA Max.)				
Mounting position	Free				
Digital interface	DeviceNet™				
Control valve actuator	Piezoelectric actuator				
Weight	1.0kg			1.4kg	

High Performance Digital Mass Flow Controller

MC-3000L Series

Outlines

- MC-3000L Series: High performance mass flow controller with high accuracy of ±1.0% S.P. guaranteed

Features

- High accuracy ±1.0% S.P.
- RS-485 digital interface
- LINTEC's unique ambient temperature-compensated sensor incorporated
- Highly functional upon installation of microprocessor
- Diaphragm valve with small dead volume
- High-speed and high performance piezoelectric actuator
- Long-term leak tightness ensured using metal seal
- By using metal case and various filters, stable operation can be achieved through reduction of radio frequency noises and electromagnetic field interferences

CE RoHS

Model		MC-3102L	MC-3202L	MC-3252L	MC-3452L	MC-3462L	MC-3472L	MC-3582L	
Flow rate in Nitrogen (Full scale)		5 to 5,000SCCM	～20SLM	～50SLM	～50SLM	～150SLM	～200SLM	～300SLM	～400SLM
Flow rate control range		2 to 100% F.S.							
Valve operation mode		Normally open, Normally closed			Normally closed				
Accuracy		100% F.S. to 25% F.S. ±1.0% S.P. 25% F.S. to 2% F.S. ±0.25% F.S.			100% F.S. to 35% F.S. ±1.0% S.P. 35% F.S. to 2% F.S. ±0.35% F.S.		±1.0% F.S.	±2.0% F.S.	±3.0% F.S.
Repeatability		±0.2% F.S.			±0.5% F.S.			±1.0% F.S.	
Response time		1 sec							2 sec
Analog flow rate setting signal		0 to 5VDC							
Analog flow rate output signal		0 to 5VDC							
Operating differential pressure		50 to 300kPa	Normally open 50 to 300kPa (～10SLM) 100 to 300kPa (～20SLM) Normally closed 100 to 300kPa (～10SLM) 200 to 300kPa (～20SLM)	100 to 300kPa (～30SLM) 150 to 300kPa (～50SLM)	150 to 500kPa	250 to 500kPa	300 to 500kPa	320 to 500kPa	350 to 500kPa
Withstand pressure		1MPa(G)							
Temperature coefficient	Zero	±0.02% F.S. / ˚C			±0.1% F.S. / ˚C				
	Span	±0.02% F.S. / ˚C	±0.04% F.S. / ˚C	±0.08% F.S. / ˚C	±0.1% F.S. / ˚C				
Operating temperature		5 to 50˚C 0 to 80% RH							
Leak integrity		1×10^{-11}Pa・m³ / sec He						1×10^{-8}Pa・m³ / sec He	
Materials exposed to gas		Stainless steel 316L, PCTFE			Stainless steel 316L, Stainless steel 304, Teflon®, Viton®				
Seal materials		Au							
Power supply requirement		+15VDC±3% 100mA, -15VDC±3% 50mA							
Mounting position		Free							
Analog connector		Dsub 9pin							
Digital interface		Modular jack 45/RS-485							
Control valve actuator		Piezoelectric actuator							
Weight		1.0kg			1.4kg			2.4kg	

Low Differential Pressure Mass Flow Controller

MC-3000S Series

Outlines

- MC-3000S Series: Mass flow controller equipped with LINTEC's original low pressure differential device to operate at low differential pressure environment

Features

- Low pressure differential device
- In addition to AsH_3 and PH_3 for ion implantation, this MFC is also applicable to solid state material and liquified gas without heating
- RS-485 digital interface
- LINTEC's unique ambient temperature-compensated sensor incorporated
- Highly functional upon installation of microprocessor
- Diaphragm valve with small dead volume
- High-speed and high performance piezoelectric actuator
- Long-term leak tightness ensured using metal seal
- By using metal case and various filters, stable operation can be achieved through reduction of radio frequency noises and electromagnetic field interferences

CE RoHS

Model		MC-3102S		
Flow rate in Nitrogen (Full scale)		2 to 10SCCM	～20SCCM	～30SCCM
Flow rate control range		2 to 100% F.S.		
Valve operation mode		Normally open, Normally closed		
Accuracy		±1.0% F.S.		
Repeatability		±0.2% F.S.		
Response time		2 sec		
Analog flow rate setting signal		0 to 5VDC		
Analog flow rate output signal		0 to 5VDC		
Operating differential pressure		8×10^2Pa to 1.33×10^5Pa	1.07×10^3Pa to 1.33×10^5Pa	1.33×10^3Pa to 1.33×10^5Pa
Withstand pressure		1MPa(G)		
Temperature coefficient	Zero	±0.02% F.S. / ℃		
	Span	±0.02% F.S. / ℃	±0.04% F.S. / ℃	±0.08% F.S. / ℃
Operating temperature		5 to 50℃ 0 to 80% RH		
Leak integrity		1×10^{-11}Pa・m^3 / sec He		
Materials exposed to gas		Stainless steel 316L, Stainless steel 304, PCTFE		
Seal materials		Au		
Power supply requirement		+15VDC±3% 100mA, -15VDC±3% 50mA		
Mounting position		Specified upon order placement		
Analog connector		Dsub 9pin		
Digital interface		Modular jack 45/RS-485		
Control valve actuator		Piezoelectric actuator		
Weight		1.0kg		

Variable Range (VR) Mass Flow Controller

MC-700 Series

Outlines

- MC-700 Series: Mass flow controller equipped with 3 variable rotary switches for gas type selection and full scale adjustment

Features

- Equipped with Variable Range (VR) function:
  VR function allows gas type and full scale range selection using "N2 default flow rate ÷ setting flow rate × conversion factor" formula, which is input by adjusting 3 rotary switches
- VR function can offer solutions to various gas type and flow rate operations and thus eliminate the need to use multiple MFCs
- Widely-used RS-485 interface equipped
  (Daisy-chain networks support up to 32 MFC connections)
- LINTEC's unique ambient temperature-compensated sensor incorporated
- Highly functional upon installation of microprocessor
- Diaphragm valve with small dead volume
- High-speed and high efficiency piezoelectric actuator
- Long-term leak tightness ensured using metal seal
- By using metal case and various filters, stable operation can be achieved through reduction of radio frequency noises and electromagnetic field interferences
- RS-232C digital interface is available as an option

CE RoHS

VR Corresponding Chart

Model		Applicable gas type	Flow rate in Nitrogen	VR value (variable)	Standard flow rate
MC-710	02	All gas types except LG	10 to 30SCCM	0.50 to 1.50	15SCCM
	03		25 to 100SCCM	0.50 to 2.00	50SCCM
	04		75 to 300SCCM	0.50 to 2.00	150SCCM
	05		0.25 to 1SLM	0.50 to 2.00	500SCCM
	06		0.75 to 3SLM	0.50 to 2.00	1.5SLM
MC-720	07		2.5 to 10SLM	0.50 to 2.00	5SLM
MC-730	08		10 to 30SLM	0.50 to 1.50	15SLM
	09		30 to 50SLM	0.60 to 1.00	30SLM

Model	MC-710	MC-720	MC-730
Flow rate in Nitrogen (Full scale)	10 to 3,000SCCM	~ 10SLM	~ 50SLM
Flow rate control range	2 to 100% F.S.		
Valve operation mode	Normally open, Normally closed		
Accuracy	±1.0% F.S.		
Repeatability	±0.2% F.S.		
Response time	1 sec		
Analog flow rate setting signal	0.1 to 5VDC		
Analog flow rate output signal	0.05 to 5VDC		
Operating differential pressure	50 to 300kPa	100 to 300kPa	150 to 300kPa
Withstand pressure	1MPa(G)		
Operating temperature	15 to 35℃ 0 to 80% RH		
Leak integrity	1×10^{-11}Pa・m^3 / sec He		
Materials exposed to gas	Stainless steel 316L, PCTFE, PTFE		
Seal materials	Stainless steel 316L		
Power supply requirement	+15VDC±3% 100mA, -15VDC±3% 50mA		
Mounting position	Free		
Analog connector	Dsub 9pin		
Digital interface	Modular jack 45/RS-485		
Control valve actuator	Piezoelectric actuator		
Weight	1.0kg		

Analog Mass Flow Controller
MC-10 Series

Outlines
- MC-10 Series: Analog mass flow controller with solenoid actuator and high variety of options available

Features
- Elastomer seal
- ±1.0% F.S. accuracy is guaranteed
- Available from 10SCCM and 50SLM
- Variable Range (VR) function is available as an option: VR function allows gas type and flow range selection using "N2 default flow rate ÷ setting flow rate × conversion factor" formula, which is input by adjusting 3 rotary switches
- VR function can offer solutions to various gas type and flow rate operations and thus eliminate the need to use multiple MFCs
- Widely used RS-485 interface is available as an option: (Daisy-chain networks support up to 32 MFC connectors)
- LINTEC's unique ambient temperature-compensated sensor incorporated
- Fast-acting solenoid actuator incorporated
- By using metal case and various filters, stable operation can be achieved through reduction of radio frequency noises and electromagnetic field interferences

CE RoHS

VR Corresponding Chart

Model		Applicable gas types	Flow rate in Nitrogen	VR value (variable)	N2 default flow rate
MC-10RC	01	All gas types except LG	10 to 20SCCM	0.50 to 1.00	10SCCM
	02		20 to 50SCCM	0.60 to 1.50	30SCCM
	03		40 to 100SCCM	0.50 to 1.25	50SCCM
	04		80 to 200SCCM	0.50 to 1.25	100SCCM
	05		200 to 500SCCM	0.60 to 1.50	300SCCM
	06		400 to 1000SCCM	0.50 to 1.25	500SCCM
	07		800 to 2000SCCM	0.50 to 1.25	1SLM
	08		2 to 5SLM	0.60 to 1.50	3SLM
MC-11RC	09		4 to 10SLM	0.50 to 1.25	5SLM
	10		8 to 20SLM	0.50 to 1.25	10SLM
	11		20 to 50SLM	0.60 to 1.50	30SLM

Model	MC-10RC	MC-11RC
Flow rate in Nitrogen (Full scale)	10 to 5,000SCCM	～50SLM
Flow rate control range	2 to 100% F.S.	
Valve operation mode	Normally closed	
Accuracy	±1.0% F.S.	
Repeatability	±0.2% F.S.	
Response time	1 sec	
Analog flow rate setting signal	0.1 to 5VDC	
Analog flow rate output signal	0 to 5VDC	
Operating differential pressure	50 to 300kPa	200 to 300kPa
Withstand pressure	1MPa(G)	
Operating temperature	15 to 35℃ 0 to 80% RH	
Leak integrity	1×10^{-8}Pa・m^3 / sec He	
Materials exposed to gas	Stainless steel 316L, Magnetic stainless steel, PTFE	
Seal materials	Viton®	
Power supply requirement	+15VDC 100mA, -15VDC 150mA	+15VDC 100mA, -15VDC 200mA
Mounting position	Free	
Analog connector	Dsub 9pin	
Control valve actuator	Solenoid actuator	
Weight	1.2kg	

High Temperature Mass Flow Controller

MC-3000L-TC Series

Outlines

- MC-3000L-TC Series: Mass flow controller with maximum operating temperature up to 120℃

Features

- Maximum operating temperature of 120℃
- RS-485 digital interface
- LINTEC's unique ambient temperature-compensated sensor incorporated
- Highly functional upon installation of microprocessor
- Diaphragm valve with small dead volume
- High-speed and high performance solenoid actuator
- Long-term leak tightness ensured using metal seal
- By using metal case and various filters, stable operation can be achieved through reduction of radio frequency noises and electromagnetic field interferences

RoHS

Model	MC-3102L-TC	MC-3202L-TC
Flow rate in Nitrogen (Full scale)	10 to 5,000SCCM	～25SLM
Flow rate control range	2 to 100% F.S.	
Valve operation mode	Normally closed	
Accuracy	±1.0% F.S.	
Repeatability	±0.2% F.S.	
Response time	1 sec	3sec
Analog flow rate setting signal	0 to 5VDC	
Analog flow rate output signal	0 to 5VDC	
Operating differential pressure	50 to 300kPa	150 to 300kPa
Withstand pressure	1MPa(G)	
Operating temperature	- L : 80 to 100℃　0 to 80% RH - H : 100 to 120℃　0 to 80% RH	
Bake temperature	150℃	
Leak integrity	1×10^{-11}Pa・m^3 / sec He	
Materials exposed to gas	Stainless steel 316L, PCTFE	
Seal materials	Au	
Power supply requirement	+15VDC±3% 100mA, -15VDC±3% 200mA	
Mounting position	Free	
Analog connector	Dsub 9pin	
Digital interface	Modular jack 45/RS-485	
Control valve actuator	Solenoid actuator	
Weight	1.2kg	

*Please consult us if you plan to apply low differential pressure, because depending on your operation conditions (gas type, temperature etc.), the MFC's flow range, response time and other specifications might change.

High Performance Digital Mass Flow Meter

MM-3000L Series

■ Outlines

- MM-3000L Series: MC-3000L Series-based highly efficient digital mass flow meter

■ Features

- RS-485 digital interface
- LINTEC's unique ambient temperature compensated sensor incorporated
- Highly functional upon installation of microprocessor
- Long-term leak tightness ensured using metal seal
- By using metal case and various filters, stable operation can be achieved through reduction of radio frequency noises and electromagnetic field interferences

CE RoHS

Model		MM-3102L	MM-3202L	MM-3252L	MM-3452L	MM-3462L	MM-3472L	MM-3582L
Flow rate in Nitrogen (Full scale)		5 to 5,000SCCM	~20SLM	~50SLM	~50SLM	~150SLM	~250SLM	~400SLM
Accuracy		100% F.S. to 25% F.S. ±1.0% S.P. 25% F.S. to 2% F.S. ±0.25% F.S.			100% F.S. to 35% F.S. ±1.0% S.P. 35% F.S. to 2% F.S. ±0.35% F.S.		±2.0% F.S.	±3.0% F.S.
Repeatability		±0.2% F.S.			±0.5% F.S.		±1.0% F.S.	
Analog flow rate output signal		0 to 5VDC						
Operating differential pressure		Less than 400kPa(D)						
Withstand pressure		1MPa(G)						
Temperature coefficient	Zero	±0.02% F.S. / ˚C			±0.1% F.S. / ˚C			
	Span	±0.02% F.S. / ˚C	±0.04% F.S. / ˚C	±0.08% F.S. / ˚C	±0.1% F.S. / ˚C			
Operating temperature		5 to 50˚C 0 to 80% RH						
Leak integrity		1×10^{-11} Pa・m^3 / sec He						1×10^{-8} Pa・m^3 / sec He
Materials exposed to gas		Stainless steel 316L			Stainless steel 316L, Teflon®, Viton®			
Seal materials		Au						
Power supply requirements		+15VDC±3% 100mA, -15VDC±3% 50mA						
Mounting position		Free						
Analog connector		Dsub 9pin						
Digital interface		Modular jack 45/RS-485						
Weight		1.0kg			1.4kg			2.4kg

High Temperature Mass Flow Meter

MM-3000L-TN Series

Outlines

- MM-3000L-TN Series: High temperature mass flow meter applicable to 120°C operation temperature

Features

- Maximum operating temperature of 120°C
- RS-485 digital interface
- LINTEC's unique ambient temperature compensated sensor incorporated
- Highly functional upon installation of microprocessor
- Long-term leak tightness ensured using metal seal
- By using metal case and various filters, stable operation can be achieved through reduction of radio frequency noises and electromagnetic field interferences

RoHS

Model	MM-3102L-TN	MM-3202L-TN
Flow rate in Nitrogen (Full scale)	10 to 5,000SCCM	～25SLM
Accuracy	±1.0% F.S.	
Repeatability	±0.2% F.S.	
Analog flow rate output signal	0 to 5VDC	
Operating differential pressure	Less than 400kPa(D)	
Withstand pressure	1MPa(G)	
Operating temperature	- L : 80 to 100°C　0 to 80% RH - H : 100 to 120°C　0 to 80% RH	
Bake temperature	150°C	
Leak integrity	1×10^{-11}Pa・m^3 / sec He	
Materials exposed to gas	Stainless steel 316L	
Seal materials	Au	
Power supply requirement	+15VDC±3% 100mA, -15VDC±3% 50mA	
Mounting position	Free	
Analog connector	Dsub 9pin	
Digital interface	Modular jack 45/RS-485	
Weight	1.2kg	

List of Connectors

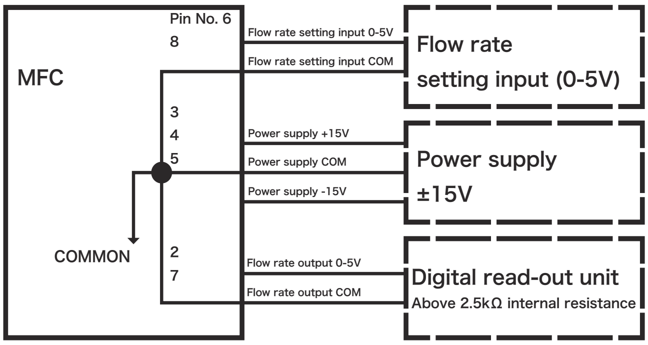
Analog Interface Connector

Connector used : D-Sub 9 pin male **Suitable connector : D-Sub 9 pin female**

Pin No.	Signal	Function
1	Input signal (valve open/close)	＋15VDC : OPEN, －15VDC : CLOSED
2	Flow rate output signal 0 to 5 VDC	Actual flow rate output voltage 0 to 5 VDC
3	Power supply input ＋15 VDC	Power supply (positive)
4	Power supply input 0 VDC	Power supply common ±15VDC
5	Power supply input －15 VDC	Power supply (negative)
6	Flow rate setting signal Hi 0 to 5VDC	Flow rate setting voltage 0 to 5 VDC
7	Flow rate output signal COMMON	Flow rate output signal COMMON
8	Flow rate setting signal COMMON	Flow rate setting signal COMMON
9	N.C.	－

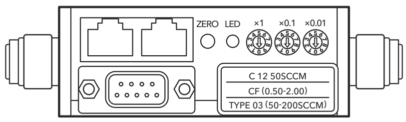
* Pin No.4, 7, 8 are connected within the apparatus.
* Impedance of valve open/close signal is 100k Ω , whereas the value of valve input resistance is equal to 100k Ω .
* Please refer to the below stated wiring instructions to eliminate COMMON potential differences.

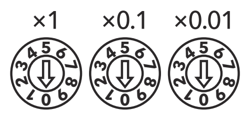
Please DO NOT connect the following to power source when high-precision control is required:
(1) Setting voltage COMMON [8]
(2) Output voltage COMMON [7]
(3) Power COMMON [4]



Variable Range (VR) Function

Variable range function allows setting different gas types and full scale ranges with a single MFC.
To change gas type or full scale range, adjust VR value of the MFC with rotary switches' or via a digital interface. To calculate the necessary VR setting, please refer to VR table or the setting formula below.





【Setting Formula】
(N2 Default Flow Rate)÷(Setting Flow Rate)×(Conversion Factor)

SCCM	SLM	01	02	03	04	05	06	07	08	09	10	11
	150											
	100											
	50											0.60
	30											
	20										0.50	1.50
	10									0.50		
	8											
	5								0.60		1.25	
	4									1.25		
	3											
	2							0.50	1.50			
1000	1						0.50	1.00				
800								1.25				
500						0.60	1.00					
400							1.25					
300						1.00						
200					0.50	1.50						
100				0.50	1.00							
80					1.25							
50			0.60	1.00								
40				1.25								
30			1.00									
20		0.50	1.50									
10		1.00										

MC-10

SCCM	SLM	02	03	04	05	06	07	08	09
	50								0.60
	30							0.50	1.00
	15							1.00	
	10						0.50	1.50	
	8								
	5						1.00		
	4								
	3					0.50			
	2.5						2.00		
1000	1				0.50				
750						2.00			
500					1.00				
400									
300				0.50					
250					2.00				
100			0.50						
75				2.00					
50			1.00						
40									
30		0.50							
25			2.00						
10		1.50							

MC-700

Product Configuration

MC - 3102L- N C - 4 VR 2 AA0A0 -(VR -)N2 - 1SLM
① ② ③ ④ ⑤ ⑥ ⑦ ⑧ ⑨ ⑩ ⑪

① MC : Mass flow Controller
MM : Mass flow Meter
② Model name
③ N : Standard treatment M : Precision treatment
④ C : Normally closed valve
O : Normally open valve
⑤ Fittings outside diameter
4 : 6.35 mm (1/4inch)
6 : 9.52 mm (3/8inch) etc.

*Please contact distributor or sales office for more details.

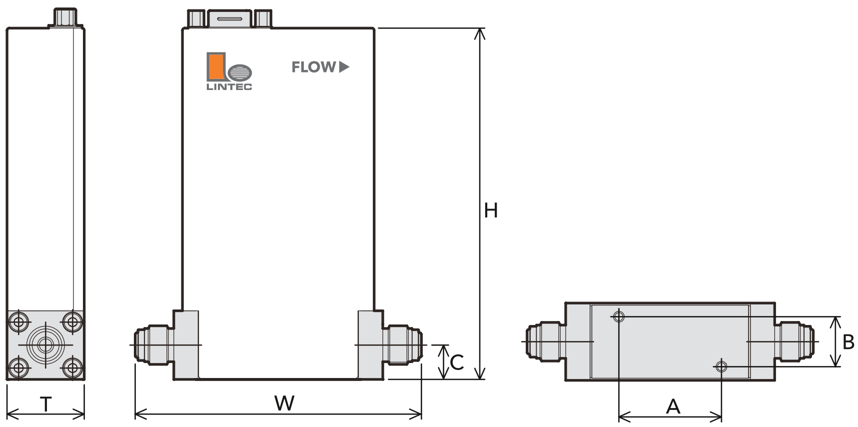
⑥ Fittings
VR : VCR SW : SWL
⑦ Width size
1 : 106 mm 2 : 124 mm
⑧ Options
(Standard : AA0A0 or A0A0A0)
⑨ VR number
(In case of MC-700 or MC-10)
⑩ Gas type
⑪ Flow rate

Dimension Table

Model	H	T	W (6.35VCR)	W (6.35SWL)	W (9.52VCR) (12.7VCR)	W (9.52SWL)	W (12.7SWL)	A	B	C
MC-7100C	137	28.6	106*1	127	-	-	-	38	18.5	12.7
MC-7200C	137	28.6	106*1	127	-	-	-	38	18.5	12.7
MC-7100L	126	28.6	106*1	127	-	-	-	38	18.5	13
MC-7200L	126	28.6	106*1	127	-	-	-	38	18.5	13
MC-5100C	137	28.6	106*1	127	-	-	-	38	18.5	12.7
MC-5200C	137	28.6	106*1	127	-	-	-	38	18.5	12.7
MC-5250C	137	28.6	106*1	127	-	-	-	38	18.5	12.7
MC-5460C	136	42	-	-	-	162.2	-	68	25.4	17
MC-5470C	136	42	-	-	-	162.2	-	68	25.4	17
MC-5580C	144	70	-	-	171.4	-	177.5	68	25.4	17
MC-5100L	126	28.6	106*1	127	-	-	-	38	18.5	13
MC-5200L	126	28.6	106*1	127	-	-	-	38	18.5	13
MC-5250L	126	28.6	106*1	127	-	-	-	38	18.5	13
MC-5460L	136	42	156	-	163.4	-	-	68	25.4	17
MC-5470L	136	42	156	-	163.4	-	-	68	25.4	17
MC-710	130	28.6	106*1	127	-	-	-	38	18.5	12.7
MC-720	130	28.6	106*1	127	-	-	-	38	18.5	12.7
MC-730	130	28.6	106*1	127	-	-	-	38	18.5	12.7
MC/MM-3102L	130	28.6	106*1	127	-	-	-	38	18.5	12.7
MC/MM-3202L	130	28.6	106*1	127	-	-	-	38	18.5	12.7
MC/MM-3252L	130	28.6	106*1	127	-	-	-	38	18.5	12.7
MC/MM-3452L	136	42	156	-	163.4	162.2	166.2	68	25.4	17
MC/MM-3462L	136	42	156	-	163.4	162.2	166.2	68	25.4	17
MC/MM-3472L	136	42	156	-	163.4	162.2	166.2	68	25.4	17
MC/MM-3582L	144	70	-	-	171.4	-	177.5	68	25.4	17
MC-3102S	130	33.8	106*1	127	-	-	-	38	18.5	12.7
MC/MM-3102L-T*2	126	32	124	127	-	-	-	38	18.5	13
MC-10RC	130	38	124	127	-	-	-	38	18.5	12.7
MC-11RC	130	38	124	127	-	-	-	38	18.5	12.7

(mm)

*1 Also available for 124mm
*2 Refers to tube length



LIQUID | Products For Liquid Applications

Featuring liquid flow rate measurement and control technologies, LINTEC's original vaporization and supply systems support advances in semiconductor manufacturing development, solar panel production lines and other high tech industries of the next generation.

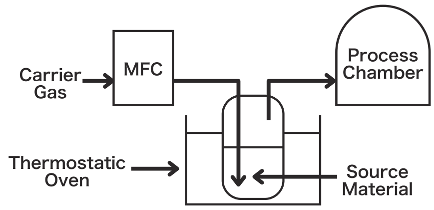
Classification of Liquid Vaporization Methods

Liquid vaporization system has come to be extensively used in pioneering semiconductor industry and other advanced manufacturing sectors. The vaporization methods are generally classified into 3 major types: (1) Bubbling method (2) Baking method, and (3) Direct vaporization method. Engaging in manufacturing products that correspond to all the stated methods, LINTEC is dedicated to provide our customers with advanced solutions that would respond to their needs the best.

[Bubbling method]

A constant amount of vapor is acquired through flow rate control of carrier gas and temperature of a liquid-containing vessel. Virtually any liquid can be vaporized utilizing this method but disadvantages include difficulty in maintaining accurate and repeatable flow measurement.

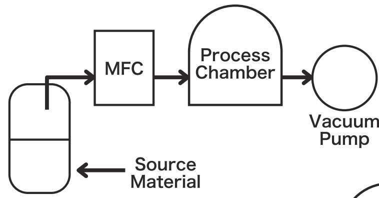
Target products: HX Series, TCU Series, and MC Series



[Baking method]

This method is conducted by operating a high temperature mass flow controller with differential pressure between the liquid material vapor pressure and the chamber pressure. Usage of a mass flow controller helps to achieve high flow rate accuracy.

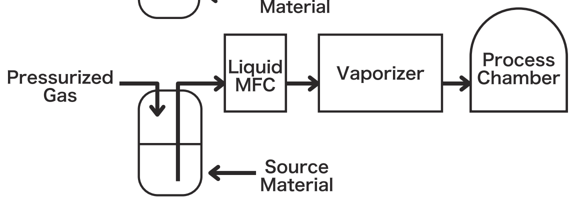
Target products: SVC Series, TCU Series, and MC Series



[Direct vaporization method]

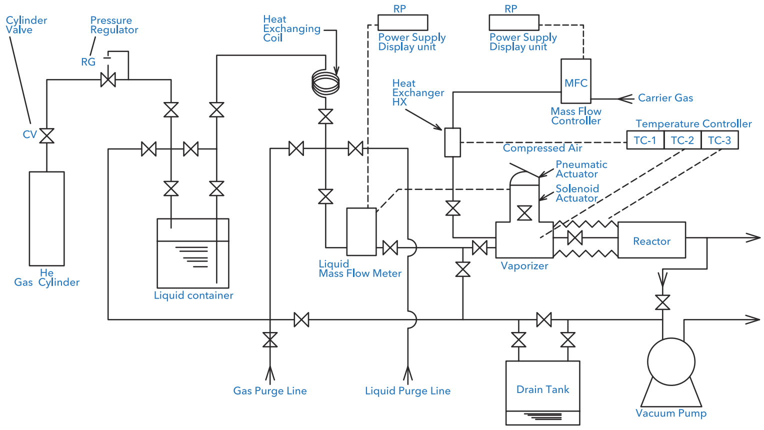
This method utilizes mass flow meter to measure precursor flow rate and vaporizer to facilitate the vaporization process. This method offers the highest efficiency for thin film deposition.

Target products: LVS Series, LM Series, VU Series, HX Series,
VF Series, TCU Series, and MC Series



Recommended Procedures for Direct Vaporization Method

Supporting delivery of large-scale vaporized liquid precursor materials, direct vaporization method has made an immense contribution to leading industries in wafer fabrication used in semiconductor, LED and solar panel manufacturing.



Thin Film Technology Using Direct Vaporization Method

Applicable film	Function/Purpose	Film material	Fluid material	Liquid MFM	Vaporizer
Insulator film	Protective film Insulator film	SiO_2 SiN PSG BPSG Others	TEOS TEB TEPO 3DMAS OMCTS TMCTS Others		VU-206 Series VU-430A Series VU-450 Series
Metallic thin film	Wiring layer Barrier metal layer Seed layer Electrode Others	Cu TiN TaN Ni Co Ru Others	TDEAT TDMAT TBTEMT DEZn TTIP $Ru(EtCp)_2$ $TiCl_4$ Others	LM-3000L Series	VU-430A Series VU-450 Series
High-k	Gate insulator Capacitor	HfO_2 ZrO_2 La_2O_3 Al_2O_3 Ta_2O_5 BST PZT SBT Others	HTB TMA TEMAH TEMAZ $Ta(OC_2H_5)_5$ $Ba(DPM)_2$ $Sr(DPM)_2$ $Pb(DPM)_2$ Others		VU-430A Series VU-450 Series VU-550 Series VU-900 Series
Others	Various applications	Various materials	H_2O Others		VU-430A Series VU-450 Series

High Performance Digital Mass Flow Controller

LC-3000L Series

Outlines

- LC-3000L Series: Highly-efficient digital mass flow controller with piezoelectric actuator incorporated within mass flow sensor for liquid applications

Features

- Operates as a liquid mass flow controller upon installation of piezoelectric actuator
- RS-485 digital interface
- LINTEC's unique ambient temperature-compensated sensor incorporated
- Highly functional upon installation of microprocessor
- Diaphragm valve with small dead volume
- High-speed and high efficiency piezoelectric actuator
- Long-term leak tightness ensured using metal seal
- By using metal case and various filters, stable operation can be achieved through reduction of radio frequency noises and electromagnetic field interferences

CE RoHS

Model		LC-3102L	LC-3702L
Flow rate	C_2H_5OH conversion	0.1 to 1.0g/min	~ 30g/min
	H_2O conversion	0.1 to 0.2g/min	~ 15g/min
Flow rate control range		5 to 100% F.S.	10 to 100% F.S.
Valve operation mode		Normally open	Normally closed
Accuracy		±1.0% F.S.	±2.0% F.S.
Response time		3sec	
Analog flow rate setting signal		0.25 to 5VDC	0.5 to 5VDC
Analog flow rate output signal		0 to 5VDC	
Operating differential pressure		50 to 300kPa	100 to 300kPa
Withstand pressure		1MPa(G)	
Operating temperature		15to 35℃ 0 to 80% RH	
Leak integrity		1×10^{-11}Pa・m^3 / sec He	
Wetted materials		Stainless steel 316L, PCTFE	Stainless steel 316L, PFA, Ni-Co
Seal materials		Au	
Power supply requirement		+15VDC±3% 100mA -15VDC±3% 50mA	
Mounting position		Free	Vertically upward towards piping or vertically downward towards the D-sub connector
Analog connector		Dsub 9pin	
Digital interface		Modular jack 45/RS-485	
Control valve actuator		Piezoelectric actuator	
Weight		1.0kg	

High Performance Digital Mass Flow Meter

LM-3000L Series

Outlines

- LM-3000L Series: High performance digital mass flow meter equipped with special sensor for liquid applications
- Operates as a liquid mass flow controller when used in combination with CV-1000 Series or control valve-fitted VU Series

Features

- Can be used as a liquid mass flow controller upon integration of external solenoid actuator control valve
- RS-485 digital interface
- LINTEC's original ambient temperature-compensated sensor incorporated
- Highly functional upon installation of microprocessor
- Long-term leak tightness ensured using metal seal
- By using metal case and various filters, stable operation can be achieved through reduction of radio frequency noises and electromagnetic field interferences

CE RoHS

Model		LM-3112L	LM-3212L	LM-3712L
Flow rate	C_2H_5OH conversion	0.1 to 1.0g/min	～5.0g/min	～30g/min
	H_2O conversion	0.1 to 0.2g/min	～1.0g/min	～15g/min
Flow rate control range		10 to 100% F.S.		
Accuracy		±1.0% F.S.		±2.0% F.S.
Analog flow rate output signal		0 to 5VDC		
Operating differential pressure		50 to 300kPa		
Withstand pressure		1MPa(G)		
Operating temperature		15 to 35℃ 0 to 80% RH		
Leak integrity		1×10^{-11}Pa・m^3 / sec He		
Wetted materials		Stainless steel 316L		
Seal materials		Au		
Power supply requirements		+15VDC±3% 100mA, -15VDC±3% 200mA		
Mounting position		Free		Vertically upward towards piping or vertically downward towards the D-sub connector
Analog connector		Dsub 9pin		
Digital interface		Modular jack 45/RS-485		
Weight		800g		

Liquid Vaporizer with Piezoelectric Actuator

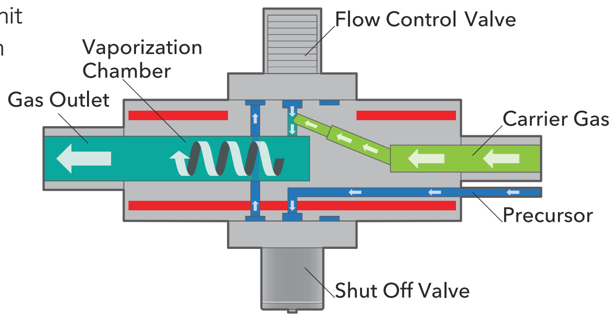
VU-206 Series

Outlines

- VU-206 Series: Piezoelectric actuator-equipped liquid vaporizer unit
- Efficient liquid vaporization control when used in combination with LM-3000L Series

Features

- Vaporization of liquid precisely controlled with piezoelectric actuator
- Maximum operating temperature up to 160˚C
- Long-term leak tightness ensured using metal seal
- Compact size

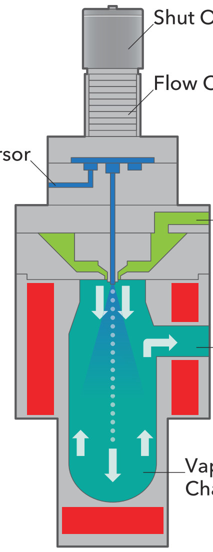


RoHS

Model		VU-0206N
Maximum vaporization rate	TEOS	5g/min
	H_2O	0.5g/min
Carrier gas flow rate		0.1 to 14SLM
Operating differential pressure		50 to 300kPa
Withstand pressure		1MPa(G)
Valve operation mode		Normally open
Maximum power consumption	120V	100W
	240V	100W
Thermocouple		K Type 1pc
Recommended temperature control method		PID
Maximum operating temperature		160˚C
Thermal switch		180˚C ±15˚C Open
Leak integrity		1×10^{-11}Pa・m^3 / sec He
Wetted materials		Stainless steel 316L, Ni-Co, Polyimide
Seal materials		Au
Control valve actuator		Piezoelectric actuator
Operating pressure of the pneumatic actuator (NC type)		500 to 600kPa(G)
Mounting position		Vertically upward towards the carrier gas inlet and actuator
Weight		1.4kg

Compact Liquid Vaporizer with Solenoid Actuator

VU-430A Series



Outlines

- VU-430A Series: Solenoid actuator equipped compact vaporizer unit
- Efficient liquid vaporization control when used in combination with LM-3000L Series

Features

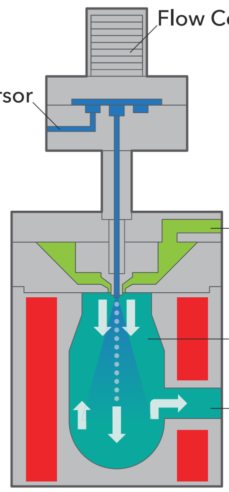
- High precision liquid vaporization control using solenoid actuator
- Compact design
- With maximum operating temperature up to 200˚C , this apparatus is most suitable for low flow rate vaporization of precursors with low vapor pressure, e.g. low-k, high-k materials and various metals
- Rapid vaporization by spraying liquid through double orifice nozzle into vaporization chamber
- Long-term leak tightness ensured using metal seal

RoHS

Model		VU-0430N
Maximum vaporization rate	TEOS	10g/min
	H_2O	1.0g/min
Carrier gas flow rate	VU-0430N-03	0.05 to 1.0SLM
	VU-0430N-02	1 to 2SLM
	VU-0430N-01	2 to 4SLM
Operating differential pressure		50 to 300kPa
Withstand puressure		1MPa(G)
Valve operation mode		Normally closed
Maximum power consumption	VU-0430N-L1	120V, 180W
	VU-0430N-L2	240V, 625W
Thermocouple		K Type 2pcs
Recommended temperature control method		PID
Maximum operating temperature		200˚C
Thermal switch		230˚C ±10˚C Open
Leak integrity		1×10^{-11}Pa・m^3 / sec He
Wetted materials		Stainless steel 316L, Ni-Co, Polyimide
Seal materials		Au
Control valve actuator		Solenoid actuator
Operating pressure of the pneumatic actuator (NC type)		500 to 600kPa(G)
Mounting position		Vertically upward towards actuator
Weight		2.2kg

Liquid Vaporizer with Solenoid Actuator

VU-450 Series



Outlines

- VU-450 Series: Vaporizer unit with Solenoid actuator
- Efficient liquid vaporization control when used with LM-3000L Series

Features

- High precision liquid vaporization system control with solenoid actuator
- This apparatus has larger vaporizing capacity than VU-430A and is most suitable for high flow rate vaporization of precursors with low vapor pressure, e.g. low-k, high-k materials and various metals
- Rapid vaporization by spraying liquid through double orifice nozzle into vaporization chamber
- Long-term leak tightness ensured using metal seal

RoHS

Model		VU-0450N	VU-0550N
Maximum vaporization rate	TEOS	20g/min	
	H_2O	5g/min	
Carrier gas flow rate	01	0.8 to 1.5SLM	
	02	1.5 to 3SLM	
	03	3 to 6SLM	
	04	6 to 10SLM	
	05	10 to 14SLM	
	06	14 to 20SLM	
	07	20 to 40SLM	
Withstand pressure		1MPa(G)	
Valve operation mode		Normally closed	
Maximum power consumption	L1	120V, 800W	
	L2	240V, 800W	
Themocouple		K Type 1pc	
Recommended temperature control method		PID	
Maximum operating temperature		200℃	300℃
Thermal switch		230℃ ±10℃ Open	350℃ ±9℃ Open
Leak integrity		1×10^{-11}Pa・m^3 / sec He	
Wetted materials		Stainless steel 316L, Ni-Co, Polyimide	
Seal materials		Au	
Control valve actuator		Solenoid actuator	
Mounting position		Vertically upward towards actuator	
Weight		6kg	

High Efficiency Liquid Vaporizer
VU-440 Series

Outlines
- VU-440 Series: Vaporizer unit with Solenoid actuator
- Efficient liquid vaporization control when used with LM-3000L Series

Features
- High precision liquid vaporization system control with solenoid actuator
- This apparatus is most suitable for high flow rate vaporization of precursors with low vapor pressure, e.g. low-k, high-k materials and various metals
- Rapid vaporization by spraying liquid through double orifice nozzle into vaporization chamber
- Long-term leak tightness ensured using metal seal
- Longer distance from the nozzle to the vapor outlet prevents any unvaporized mist from going into process chamber

RoHS

Model		VU-0440N
Maximum vaporization rate	TEOS	30g/min
	H_2O	5g/min
Carrier gas flow rate	01	0.8 to 1.5SLM
	02	1.5 to 3SLM
	03	3 to 6SLM
	04	6 to 10SLM
	05	10 to 14SLM
	06	14 to 20SLM
	07	20 to 40SLM
Withstand pressure		1MPa(G)
Valve operation mode (for a valve-equipped model)		Normally closed
Maximum power consumption		240V, 960W
Themocouple		K Type 1pc
Recommended temperature control method		PID
Maximum operating temperature		200℃
Thermal switch		230℃ ±10℃ Open
Leak integrity		5×10^{-6}Pa・m^3 / sec He
Wetted materials		Stainless steel 316L, Ni-Co, Polyimide
Seal materials		Au, Karlez®
Mounting position		Vertically upward towards actuator
Weight		9kg

Vaporizer with Selectable Actuator Type

VU-462 Series

Outlines

- VU-462 Series: Vaporizer with solenoid or piezoelectric selectable actuator type. Valveless version is also available.
- Efficient liquid vaporization control when used with LM-3000L Series

Features

- Actuator type selection allows users to build a customized vaporization supply system optimized for a specific application
- This apparatus has a larger vaporizing capacity than VU-450 and is most suitable for high flow rate vaporization of precursors with low vapor pressure, e.g. low-k, high-k materials and various metals
- Rapid vaporization by spraying liquid through double orifice nozzle into vaporization chamber
- Long-term leak tightness ensured using metal seal
- Longer distance from the nozzle to the vapor outlet prevents any unvaporized mist from going into process chamber

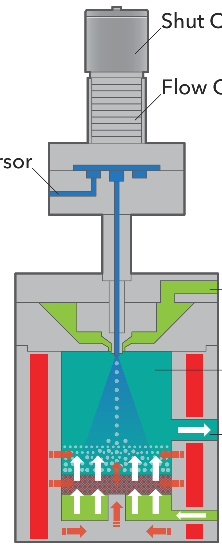
RoHS

Model		VU-0462N-T23	VU-0462N-T35
Maximum vaporization rate	TEOS	50g/min	
	H_2O	15g/min	
Carrier gas flow rate	01	0.8 to 1.5SLM	
	02	1.5 to 3SLM	
	03	3 to 6SLM	
	04	6 to 10SLM	
	05	10 to 14SLM	
	06	14 to 20SLM	
	07	20 to 40SLM	
Withstand pressure		1MPa(G)	
Valve operation mode (for a valve-equipped model)		Normally closed	
Maximum power consumption		240V, 2400W	
Thermocouple		K Type 1pc	
Recommended temperature control method		PID	
Maximum operating temperature		200℃	300℃
Thermal switch		230℃ ±10℃ Open	350℃ ±9℃ Open
Leak integrity		5×10^{-6}Pa・m^3 / sec He	5×10^{-10}Pa・m^3 / sec He
Wetted materials		Stainless steel 316L, Ni-Co, Polyimide	
Seal materials		Au, Karlez®	Au
Mounting position		Vertically upward towards actuator	
Weight		9kg	

Light-Contact Vaporizer
VU-900 Series

Outlines
- VU-900 Series: Vaporizer providing stable vaporization of liquid source by preventing heat deterioration through reducing possibility of direct contact of liquid with vaporization chamber
- Efficient liquid vaporization control when used with LM-3000L Series

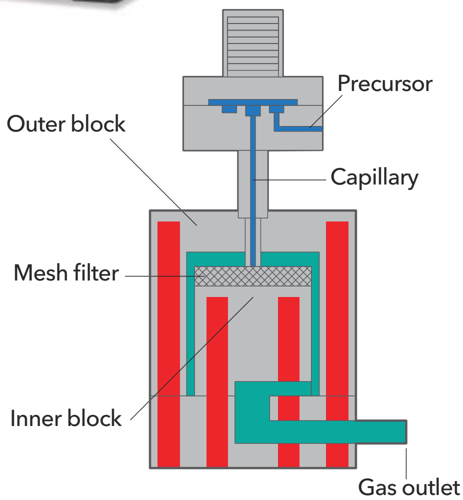
Features
- High precision liquid vaporization system control using solenoid actuator
- Most suitable for thermally unstable materials such as low-k, high-k, and various metals
- Rapid vaporization by spraying liquid through double orifice nozzle into vaporization chamber
- Long-term leak tightness ensured using metal seal



RoHS

Model		VU-0920N
Maximum vaporization rate	H_2O	5g/min
Operating differential pressure		50 to 300kPa
Withstand pressure		1MPa(G)
Valve operation mode		Normally closed
Maximum power consumption	120V	800W
	240V	800W
Thermocouple		K Type 1pc
Recommended temperature control method		PID
Maximum operating temperature		300℃
Thermal switch		350℃ ±9℃ Open
Leak integrity		1×10^{-11}Pa・m^3 / sec He
Wetted materials		Stainless steel 316L, Ni-Co, Polyimide
Seal materials		Au
Control valve actuator		Solenoid actuator
Mounting position		Upward towards connector and horizontal towards piping
Weight		5kg

Non-Carrier Gas Vaporizer
VU-3000 Series



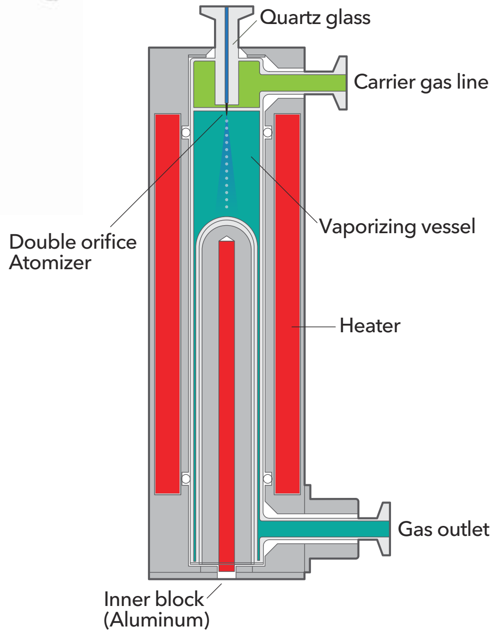
■ Outlines
- VU-3000 Series: Vaporizing multi-precursor without carrier gas
- Efficient liquid vaporization control when used with LM-3000L Series

■ Features
- High precision liquid vaporization system control with solenoid actuator
- Has maximum operating temperature of 200℃ , compact design and is suitable for many precursor types
- Upon entering the vaporizer the liquid is diffused via a mesh filter, which further enhances vaporization efficiency
- Long-term leak tightness ensured using metal seal

RoHS

Model		VU-3100N	VU-3000
Maximum vaporization rate	H_2O	-	5g/min
	TEOS	5g/min	-
Operating differential pressure		50 to 300kPa	250 to 300kPa
Withstand pressure		1MPa(G)	
Valve operation mode		Normally open	Normally closed
Maximum power consumption	120V	150W	Inner: 625W Outer: 417W
	240V	150W	Inner: 900W Outer: 600W
Thermocouple		K Type 1pcs	K Type 2pcs
Recommended temperature control method		PID	
Maximum operating temperature		150℃	200℃
Thermal switch		180℃ ±10℃ Open	270℃ ±10℃ Open
Leak integrity		5×10^{-10}Pa・m^3 / sec He	5×10^{-6}Pa・m^3 / sec He
Wetted materials		Stainless steel 316L, PEEK®, Ni-Co, PTFE, Au	Stainless steel316L, Karlez®, Polyimide, Au
Mounting position		Upward towards connector and horizontal towards piping	Vertically upward towards the air valve and horizontal towards piping
Weight		600g	5kg

Quartz Vaporizer
QV Series



Outlines
- QV-1000 Series: Metal-free vaporizer made with quartz glass. Enables vaporization of corrosive precursors without risking metal contamination

Features
- Metal-free construction utilizing quarts glass for wetted parts
- Available for precursors that would cause corrosion to stainless steel, e.g. H_2O_2
- Precursor is atomized and supplied to the vaporization chamber via a double orifice nozzle and then is vaporized by passing in between the inner and outer quartz glass heater blocks which provide highly efficient vaporization
- Guaranteed vaporization rate of 15g/min H_2O

RoHS

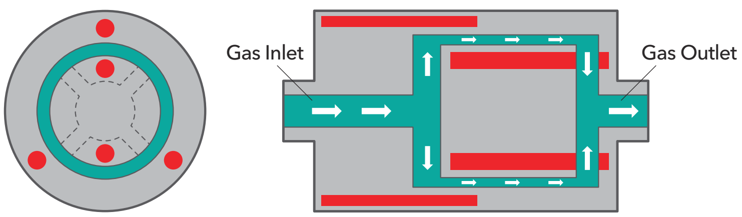
Model		QV Series
Maximum vaporization rate	H_2O	15g/min
Carrier gas flow rate		N_2 10SLM
Withstand pressure		200kPa･A
Maximum power consumption	240V	3400W
Thermocouple		K Type 2pcs
Recommended temperature control method		PID
Maximum operating temperature		200℃
Thermal switch		Inner block: 270℃ ±10℃ Open Outer block: 270℃ ±10℃ Open
Leak integrity		5×10^{-4}Pa・m^3 / sec He
Wetted materials		Quartz glass
Fittings		NW16 Glass Flange
Mounting position		Vertically upward towards the precursor inlet and horizontal towards piping
Weight		12kg

OTHERS

Pioneering the development of gas and liquid vaporization systems, LINTEC provides our customers with a variety of products besides mass flow controllers. By creating sophisticated products and providing low-carbon solutions to our society, we strive to become an innovation-driven company that makes the impossible possible.

Heat Exchanger
HX Series





[HX Inner Construction]

■ Outlines

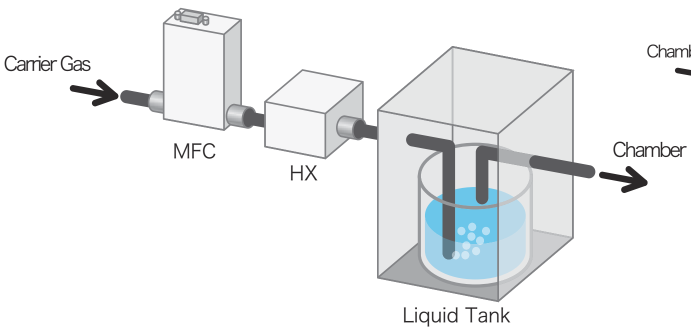
- HX Series: Compact heat exchanger utilizing LINTEC's high-efficiency vaporization techniques

■ Features

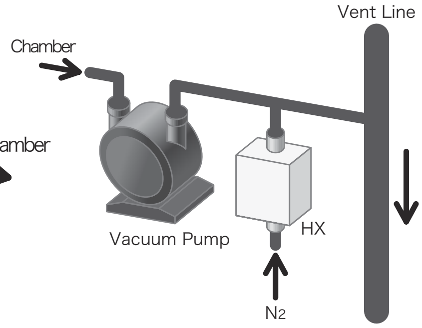
- Heatable up to maximum 300℃
- LINTEC's original design remarkable for high themal efficiency, low pressure loss and clean inner strucuture
- Metal-composed, clean casing
- Thermal switch for aberrant heating detection installed

RoHS

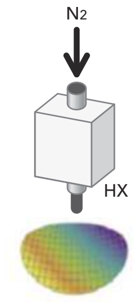
■ Application examples



(1)Carrier Gas Heating



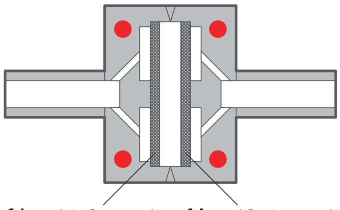
(2)Exhaust Gas Heating



(3)Wafer Distortion Correction

Model		HX-0020	HX-0050A	HX-0051H	HX-0100C	HX-0300C
Flow rate in Nitrogen		10SLM	50SLM	20SLM	100SLM	300SLM
Pressure loss		14kPa (10SLM N_2)	9.8kPa (50SLM N_2)	40kPa(20SLM N_2)	9.8kPa (100SLM N_2)	30kPa (300SLM N_2)
Withstand pressure		1MPa(G)				
Maximum power consumption	120V	100W	500W	400W	700W	-
	240V	100W	875W	400W	875W	2100W
Thermocouple		K Type 1pc			K Type 2pcs	
Recommended temperature control method		PID				
Maximum operating temperature		200℃		300℃		250℃
Thermal switch		230℃ ±10℃ Open		350℃ ±20℃ Open	350℃ ±9℃ Open	
Leak integrity		1×10^{-11}Pa・m^3 / sec He				
Materials exposed to gas		Stainless steel 316L				
Mounting position		Free				
Weight		1.0kg	1.6kg	2.0kg	5.5kg	

Heating Filter
VF Series



■ Outlines
- VF Series: Heating filter set at the back of a vaporization system for particles and mist elimination

■ Features
- Filtration accuracy of 0.1 μm
- Heatable up to 200℃
- Metal-composed, clean casing
- Thermal switch for aberrant heating detection installed

RoHS

Model		VF-0201N
Pressure loss		1.33kPa (500SCCM N_2)
Withstand pressure		1MPa(G)
Maximum power consumption	120V	400W
	240V	400W
Thermocouple		K Type 1pc
Recommended temperature control method		PID
Maximum operating temperature		200℃
Thermal switch		230℃ ±10℃ Open
Leak integrity		1×10^{-11} Pa・m^3 / sec He
Materials exposed to gas		Stainless steel 316L
Mounting position		Free
Weight		4.0kg

Temperature Control Unit
TCU Series

■ Outlines
- TCU Series: Temperature control unit for vaporizer, heat exchanger, supplying tank and pipe heater

■ Features
- Electric thermoregulator and SSR integrated
- Auto power-off function when unusual rise in temperature is detected

RoHS

Model	TCU-3-1	TCU-3-2	TCU-3-2S
Temperature control	1 system	2 independent systems	2 interconnected systems
Rated voltage	AC100 to 240V 50/60Hz		
Maximum power consumption	33VA	40VA	
Operating temperature range	0 to 50℃ 25 to 70% RH		
Weight	3.5kg	3.7kg	
Target products	HX-0050A, HX-0020, VU-0206N, VU-0450N, VF-0201N		HX-0100C HX-0300C VU-0430N VU-3000N

Piezoelectric Actuator Control Valve

CV-3000 Series

■ Outlines

- CV-3000 Series: Pressure-control, high-speed-reaction valve that controls flow rate and concentration of gas

■ Features

- High-speed and high efficiency piezoelectric actuator
- Long-term leak tightness ensured using metal seal
- By using metal case and various filters, stable operation can be achieved through reduction of radio frequency noises and electromagnetic field interferences
- Applicable to both open and closed loops

Model	CV-3102	CV-3202
Flow rate in Nitrogen	100CCM, 500CCM, 5LM	10LM
Valve operation mode	Normally open, normally closed	
Flow rate control range	2% highest flow rate	
Valve aperture setting	0 to 5VDC	
Response time	Less than 0.5 sec	
Operating differential pressure	50 to 300kPa	Normally open 50 to 300kPa Normally closed 100 to 300kPa
Maximum operating differential pressure	300kPa(G)	
Withstand pressure	1MPa(G)	
Operating temperature	0 to 50℃ · 0 to 80%RH	
Leak integrity	1×10^{-11}Pa · m^3 / sec He	
Materials exposed to gas	Stainless steel 316L, PCTFE, Au	
Seal materials	Au	
Power supply requirement	+15VDC±3% 50mA, -15VDC±3% 50mA	
Mounting position	Free	
Control valve actuator	Piezoelectric actuator	
Weight	700g	

Solenoid Actuator Control Valve (Liquid)

CV-1000 Series

■ Outlines

- CV-1000 Series: Solenoid actuator-equipped control valve for liquid applications
- Efficient liquid control when used in combination with LM-3000L Series

■ Features

- High-speed and high efficiency solenoid actuator
- Long-term leak tightness ensured using metal seal
- By using metal case and various filters, stable operation can be achieved through reduction of radio frequency noises and electromagnetic field interferences

Model		CV-1204	
Flow rate	C_2H_5OH conversion	0.1 to 10g/min	10 to 20g/min
	H_2O conversion	0.1 to 5g/min	5 to 10g/min
Flow rate control range *		5 to 100% F.S.	
Valve operation mode		Normally closed	
Accuracy *		±1.0% F.S.	
Linearity *		±0.5% F.S.	
Repeatability *		±0.5% F.S.	
Response time *		6sec	
Analog flow rate setting signal *		0 to 5VDC	
Analog flow rate output signal *		0 to 5VDC	
Operating differential pressure		50 to 300kPa	100 to 300kPa
Withstand pressure		1MPa(G)	
Operating temperature		0 to 50℃ 0 to 80% RH	
Leak integrity		1×10^{-11}Pa · m^3 / sec He	
Wetted materials		Stainless steel 316L, PTFE	
Seal materials		Au	
Power supply requirement		+15VDC±3% 100mA, -15VDC±3% 200mA	
Mounting position		Free	
Control valve actuator		Solenoid actuator	
Weight		750g	

*When used with LM-3000L

Automatic Liquid Supply System

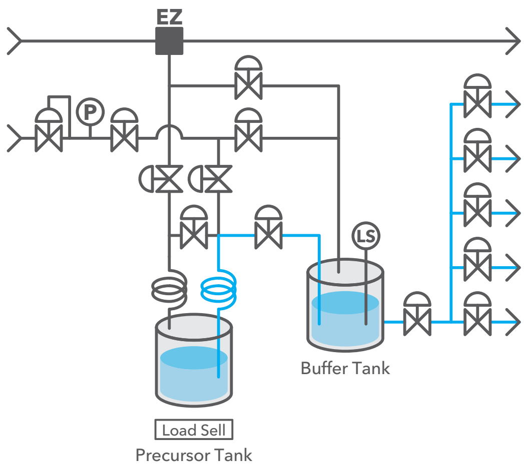
LSS Series

■ Outlines

- LSS Series: Automatic liquid supply apparatus designed to supply liquid from the inner tank to multiple tanks

■ Features

- Supports multiple/individual supply lines
- Clean design makes maintenance convenient
- Various alarm functions equipped
- Various auto-purge functions equipped
- Safe and stable design optimal for various precursors



Liquid Vaporization System

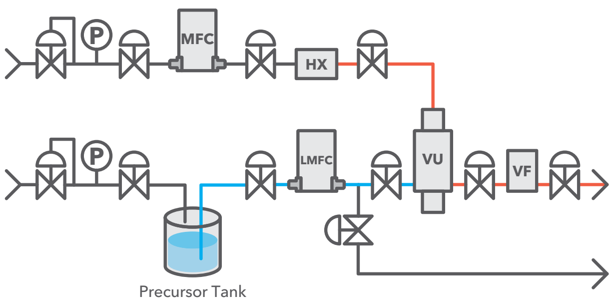
LVS Series

■ Outlines

- LVS Series: Compact-size vapor delivery system utilizing both liquid mass flow meters, controllers and vaporizers

■ Features

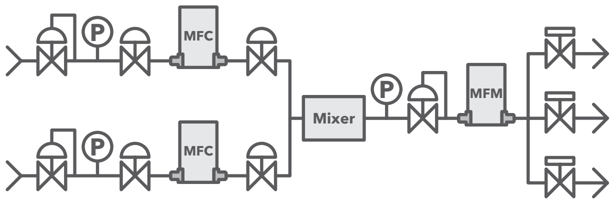
- All-in-one design is compatible with wiring of various liquid vapor supply utilities
- Customization depends on physical properties of the precursor, flow rate and operating environment
- Clean design makes maintenance convenient
- Various alarm functions equipped
- Safe and stable design optimal for various precursors



Multi Gas Mixer
FL Series

■ Outlines
- FL Series: Multi gas mixer supplies multiple gas types with specified flow rates and concentration utilizing mass flow controller

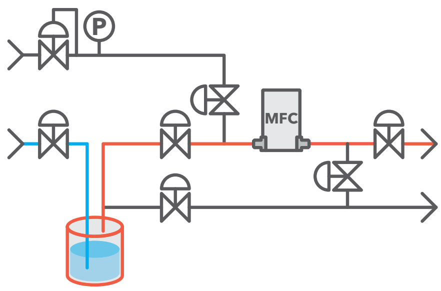
■ Functions
- Appropriately administering specific amount of gas mixture in accordance with the customer's needs
- Automatically supplying gas mixtures with constant concentrations without being affected by the concentrations of the gas supplied
- Customization depends on physical properties of precursor, flow rate and operating environment
- Clean design makes maintenance convenient
- Various alarm functions equipped
- Safe and stable design optimal for various precursors



Baking Unit
SVC Series

■ Outlines
- SVC Series: Baking unit delivers vaporized gas supply through a mass flow controller within an incorporated tank

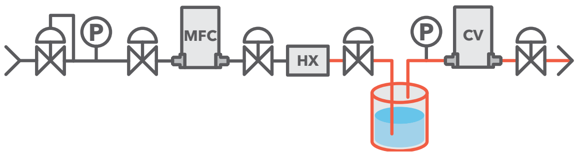
■ Functions
- Optimizing temperature control of the supply line to avoid liquefaction
- All-in-one design is compatible with wiring of various bubbling supply utilities
- Customization depends on physical properties of precursor, flow rate and operating environment
- Clean design makes maintenance convenient
- Various alarm functions equipped
- Safe and stable design optimal for various precursors



Bubbling Unit
BC Series

■ Outlines
- BC Series: Flows carrier gas into precursor bubbling tank and supplies precursor after vaporization

■ Features
- Heating up carrier gas to prevent liquefaction in supply line after controlling flow rate with MFC
- Customization depends on physical properties of precursor, flow rate and operating environment
- All-in-one design is compatible with wiring of various bubbling supply utilities
- Safe and stable design optional for various precursors
- Clean design makes maintenance convenient



Gas Panel
GAS PANEL Series

■ Outlines
- GAS PANEL Series: Integrates MFCs, filters, valves etc. to supply the necessary amount of gas with more stability

■ Features
- Applicable for various fitting types（SWL, VCR, surface mount etc.）
- Customization depends on physical properties of precursor, flow rate and operating environment
- For safety concerns, it can be equipped with another alarm or sensor device
- Clean design makes maintenance convenient

Power Supply/Read-Out Unit for MFC

RP-310

■ Outlines

● This unit is a power supply which designed for display, setting of LINTEC's mass flow controller.

■ Features

● MFC power supply, setting and display all-in-one, one touch button to control the MFC.
● The display range can be adjusted according to the F.S. of the connected MFC, it's possible to directly read the output.
● Support external control, forced open/close
● With MFC totalizer function, flowrare setting/output switch.
● Lockable switch to avoid operation errors
● Din size specification 48mm x 96mm
● Wide range of rated voltage between AC100~240V

CE RoHS

Model	RP-310
Measurement contents	Ratemeter / Totalizer
Measurement accuracy	±0.3% (F.S.) ± 1digit for analog input (23℃)
Power supply requirement	AC100-240V (-15% / +10%) 270mA max 50/60Hz less than 35VA
Setting voltage output for MFC	0 to 5VDC 5mA D-Sub connector (10 turn front panel potentiometer)
Measurement of MFC Output	Measurement range: 0 to 5VDC (A/D conversion operation) Input impedance: 330 k Ω Input measurement interval: 20ms Display sampling: every 0.5sec
Display scaling (Max. displayed digits)	Can be set to values 0.001 to 9999 during 5VDC input
External signal	Command input: 0 to 5VDC Input impedance: 330 k Ω Valve override ±15VDC
Operating conditions	0 to 50℃ 30 to 80% RH (no condensation)
EMC Directives (CE)	EN61326-1、EN61010-1、EN50581

Power Supply Unit for MFC

PS Series

■ Outlines

● PS Series: Power supply unit required to operate mass flow controllers and mass flow meters

■ Features

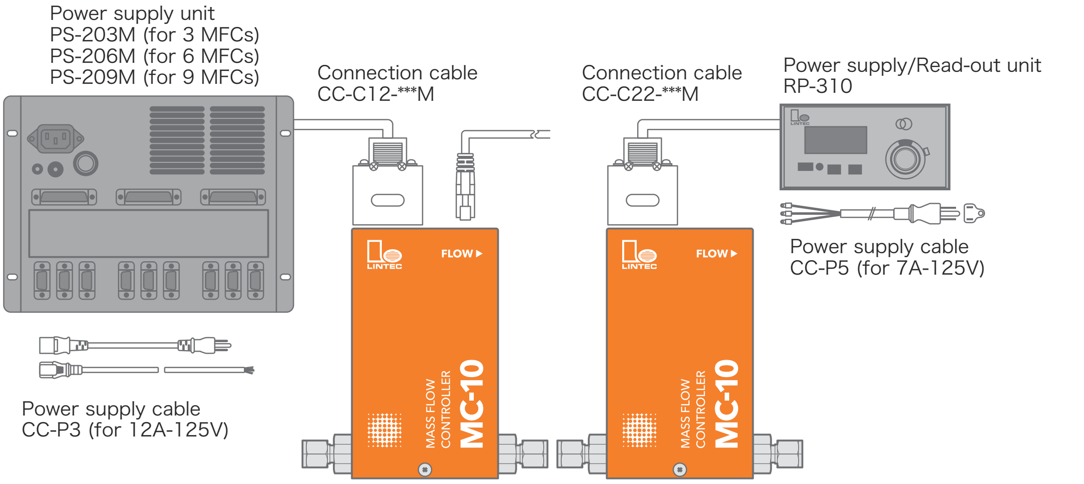
● Wide range of rated voltage between AC100~240V
● Supported external control

CE RoHS

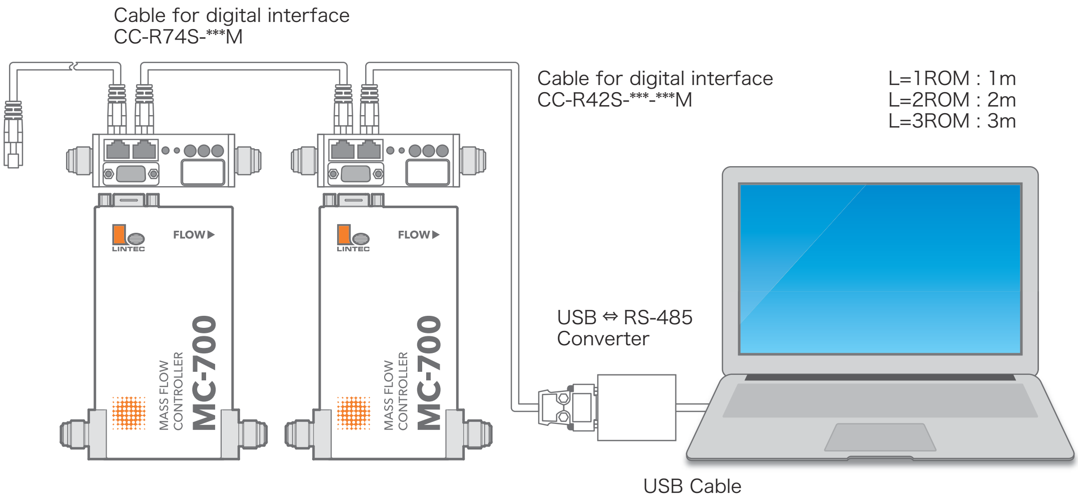
Model		PS-203M	PS-206M	PS-209M
Power supply requirement		AC85 to 240V 115W		
Setting power input		3 Channels 0 to 5 VDC (Input impedance 10M Ω or more)	6 Channels 0 to 5 VDC (Input impedance 10M Ω or more)	9 Channels 0 to 5 VDC (Input impedance 10M Ω or more)
Valve override input		3 Channels ±15VDC	6 Channels ±15VDC	9 Channels ±15VDC
Power output for MFC		±15V, 0.6 A	±15V, 1.2 A	±15V, 2.0 A
Power output for indicator		+5V, 1.0 A	+5V, 1.2 A	+5V, 2.0 A
Standard voltage output		3 Channels 5 VDC (Max. of each channel 2mA)	6 Channels 5 VDC (Max. of each channel 2mA)	9 Channels 5 VDC (Max. of each channel 2mA)
MFC flow rate output		3 Channels 0 to 5 VDC	6 Channels 0 to 5 VDC	9 Channels 0 to 5 VDC
Connector	MFC	Dsub 9pin(Female) 3pcs	Dsub 9pin(Female) 6pcs	Dsub 9pin(Female) 9pcs
	Indication setting	Dsub 15pin(Female) 1pc	Dsub 15pin(Female) 2pcs	Dsub 15pin(Female) 3pcs
Operating conditions		5 to 45℃ 0 to 85% RH		

Connection Diagram for Appurtenances and Parts

[Power supply only]　　[Power supply/Read-out Unit 2 in 1]



[Connection Diagram for Digital Interface]



[Connection Diagram for Temperature Control Unit]